

Research on Scenario-Specific Evaluation of Converter Equipment Operation and Maintenance and Intelligent Agent Decision-Making Technology for Operational Maintenance

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Abstract—Objective: To address the challenges of complex operating conditions, diverse fault types, and lagging traditional maintenance strategies in converter equipment, this study proposes a machine learning-based approach combining scenario-specific evaluation with agent-based maintenance decision-making. **Method:** A TCN+Transformer multi-task diagnostic model was constructed to extract temporal features, integrated with a PPO reinforcement learning algorithm to optimize dynamic maintenance strategies. **Results:** Empirical analysis demonstrates that the model achieves a Macro-F1 score of 89.9% in scenario recognition and an AUC of 95.4% in fault diagnosis. Over a 30-day period, comprehensive maintenance benefits increase to 2.09 million yuan, availability rises to 98.2%, and maintenance costs decrease by 19%. **Conclusion:** The proposed model effectively enables intelligent state recognition and cost-efficient operation of converter equipment, providing reliable support for the safe operation of flexible DC systems.

Keywords—Converter equipment; Sub-scenario evaluation; Agent decision-making; Machine learning

I. INTRODUCTION

Facing the fast development of flexible DC transmission, the safe and reliable operation of converter equipment have increasingly become the main limiting factor of system reliability. The O&M mode depending on experience and rules developed for conventional O&M systems is difficult to meet the challenge posed by the complex and dynamic operating conditions and equipment features of aging. Thus, this paper proposes a machine learning based scenario segmentation evaluation method considering the multi-source temporal data characteristics of O&M for converter equipment. An agent based O&M decision-making model is established to realize the comprehensive state awareness, fault diagnosis and optimal scheduling. The research results of this paper provide theoretical basis and engineering reference for intelligent and autonomous O&M for converter equipment^[1].

II. THEORETICAL ANALYSIS

A. Characteristics and Evaluation Mechanism of Subdivided Scenarios for Converter Equipment Operation and Maintenance

Because the operational status of converter equipment is the coupled effect of many kinds of factors, such as temperature,

voltage, current and cooling, the temporal characteristic of its nonlinear and highly random.

In order to describe the evolution characteristics of health status, this paper constructs multi-source feature vectors and defines Health Index (HI) to represent equipment degradation degree from all aspects. Further, TCN and Transformer networks are integrated to extract deep temporal features for multi-task scenario classification, fault identification and lifetime prediction. The proposed evaluation model can accurately identify the possible abnormalities of equipment under complex operating conditions and offers quantitative support for the following agent^[2-5].

B. Theoretical Foundation of Intelligent Agent Maintenance Decision-Making

To solve the dynamic decision-making problem of converter equipment maintenance, this paper generalizes maintenance behaviors into a Partially Observable Markov Decision Process (POMDP). Through the definition of states, actions and reward functions, an environment with equipment health, maintenance cost and availability as expected targets is established. Further, Policy Proportional Optimization (PPO) algorithm tunes the policy network to learn the reinforcement learning algorithm, and allows the agent to explore the optimal maintenance strategy through self-learning in a simulation environment. Finally, the proposed solution improves equipment availability^[6-8].

III. MODEL CONSTRUCTION AND ALGORITHM DESIGN

A. Construction of the Detailed Scenario Evaluation Model

To achieve multidimensional feature extraction and scenario recognition for the complex operational states of converter equipment, this paper constructs a detailed scenario evaluation model based on the fusion of Time-Convolutional Networks (TCN) and Attention Mechanisms (Transformer). The model takes multi-source time-series vectors $X_{t-L:t}$ as input, undergoes deep feature mapping, and outputs a probability distribution of scenario categories. This model simultaneously optimizes classification and health prediction tasks through multi-task learning to enhance generalization capability and interpretability. The calculation formula is as follows.

$$\hat{y}_t = \text{softmax}(w_2 \sigma(w_1 \phi(X_{t-L:t}) + b_1) + b_2) \quad (1)$$

Here, $\phi(\cdot)$ denotes the TCN-Transformer feature extraction function, $\hat{y}_t$ representing the prediction results for subdivided scenes.

B. Design of Intelligent Agent Operations and Maintenance Decision Model

For the dynamic decision-making problem in equipment operation and maintenance, this paper introduces an agent model based on Proximal Policy Optimization (PPO) reinforcement learning. Through state-action-reward interactions, the agent optimizes the maintenance policy. Given the observed equipment state o_t , the agent selects an action a_t to maximize the long-term expected return, as expressed by the following formula.

$$\pi_{\theta}^*(a_t | o_t) = \arg \max_{\pi_{\theta}} E \left[\sum_{t=0}^T \gamma^t R_t \right] \quad (2)$$

Here, π_{θ} denotes the policy function, γ represents the discount factor, and R_t refers to the instantaneous reward function.

C. Model Coupling and System Integration

To achieve collaborative optimization of converter equipment condition recognition and O&M decision-making, this paper constructs a “diagnosis-decision-feedback” closed-loop system by coupling a scenario-specific evaluation model with an agent-based O&M decision model. The scenario-specific evaluation model outputs equipment health features and scenario labels based on multi-source monitoring data, providing agents with high-dimensional, low-noise state representations. The agent dynamically selects optimal maintenance actions based on status information and historical O&M experience, achieving adaptive strategy updates and benefit maximization. The system integration layer employs a joint optimization mechanism to enhance classification accuracy and strategy stability. The model accurately identifies equipment status and enables optimal maintenance and scheduling decisions under complex operating conditions, improving O&M intelligence and economic efficiency. During deployment environment testing, the model achieved an average inference time of 84.6ms, meeting real-time requirements. Running on an Intel i7-11800H + RTX 3060 platform, it exhibited approximately 42% CPU utilization, under 1.2GB of GPU memory usage, and power consumption controlled below 65W, demonstrating robust edge deployment capabilities^[9-11].

IV. SIMULATION EXPERIMENTS AND ANALYSIS

A. Simulation Experiment Design

To validate the proposed model's effectiveness and generalization capability under complex operating conditions, this paper constructs an operation and maintenance simulation platform based on a ± 500 kV modular multilevel converter (MMC). The system operates at a sampling frequency of 1 Hz, collecting key parameters including valve temperature, arm current, submodule voltage, cooling temperature difference,

and ambient temperature. Four typical operational scenarios were established: normal operation, minor anomalies, planned maintenance, and pre-failure indications. Scenario recognition was achieved by inputting multi-source time-series signals into the TCN-Transformer model, with results fed back to the reinforcement learning agent for strategy optimization. The dynamic evolution of system health status over time can be simplified as follows.

$$HI_{t+1} = HI_t + \alpha \Delta t + \beta \sqrt{\Delta t} \epsilon_t - \gamma(a_t) \quad (3)$$

Here, HI_t denotes the equipment health index; α represents the natural degradation rate; β is the stochastic disturbance coefficient; $\epsilon_t \sim N(0,1)$; $\gamma(a_t)$ denotes the health recovery induced by the maintenance action. This model is employed to simulate the variations in health status under different maintenance strategies, thereby evaluating the long-term returns and availability performance of the agent's policy. In addition, to analyze the generalization ability of the model across different voltage levels, the MMC converters with ± 320 kV and ± 800 kV were tested in a cross-level application. When the voltage level was ± 320 kV, the model achieved a Macro-F1 score of 88.5% and an AUC of 94.2%. At ± 800 kV, the Macro-F1 score further increased to 90.3% and the AUC reached 95.1%. These results indicate that the model exhibits a certain degree of generalization across voltage levels and demonstrates good scalability.

B. Experimental Results and Analysis

To compare the performance of different models in granular scenario recognition and intelligent operations and maintenance decision-making, this paper selects four methods for comparative analysis: the traditional rule-based method (Rule), the temporal convolutional network (TCN-only), the LSTM-DQN model, and the proposed fusion model (Ours). Experimental results are reported as test set averages, with metrics including Macro-F1 (scenario recognition accuracy), AUC (fault diagnosis performance), comprehensive O&M benefit J, availability improvement ΔA , and cost reduction ΔC .

Table 1. Comparison of Experimental Results for Different Models

Method	Macro-F1 (%)	AUC (%)	Return J ($\times 10^4$ CNY)	ΔA (percentage points)	ΔC (%)
Rule	78.6	85.2	112	0	0
TCN-only	86.9	90.7	148.3	1.9	8
LSTM-DQN	88.1	91.5	171.2	2.6	12
Ours (TCN + Transformer + PPO)	92.7	95.4	209	4.1	19

Table 1 shows the performance of different methods. Our TCN+Transformer multi-task diagnosis and PPO agent decision model is better than other comparison methods on all metrics. Specifically, Macro-F1 improves by 14.1 percentage points and AUC improves by 10.2 percentage points than Rule model, which means our model can recognize the scenario more accurately and the fault type is more accurate. Comprehensive operational benefit J improves about 866,000 yuan in total. Our agent comprehensively considers the maintenance cost and the availability of equipment in the long-term operation and the economic effect is better. Availability improves about 4.1

percentage points and the maintenance cost decreases about 19%, which validates that our model is stable and applicable in practical scenarios with complex operating conditions^[12].

V. EXPERIMENTAL RESULTS METRICS EVALUATION

A. Scenario Evaluation Performance Metric (Macro-F1)

To comprehensively evaluate the classification performance of the recognition model for specific operational scenarios of converter equipment maintenance, this paper adopts the Macro-F1 score as the primary metric. This metric balances precision and recall, mitigates bias caused by imbalanced scene samples, and better reflects the model's overall discrimination capability across various operational states. The Macro-F1 score is defined as follows.

$$\text{Macro-F1} = \frac{1}{K} \sum_{k=1}^K \frac{2 \cdot P_k \cdot R_k}{P_k + R_k} \quad (4)$$

Here, K denotes the total number of sub-scenario categories (in this study, $K=4$, i.e., S0–S3), while P_k and R_k represent the precision and recall rates for category k , respectively. This metric ranges from $[0,1]$, with higher values indicating superior model recognition performance.

To validate the classification capabilities of the TCN+Transformer multi-task learning model constructed in this paper across different scenarios, we statistically analyzed the classification results for four categories of scenarios on the test set, as shown in Table 2.

Table 2. Classification Results and Macro-F1 Statistics Across Different Scenarios

Scenario Category	Precision P (%)	Recall R (%)	F1 Score (%)	Sample Proportion (%)
S0 Normal Operation	95.3	93.8	94.5	46.2
S1 Minor Anomaly	91.2	89.6	90.4	28.4
S2 Scheduled Maintenance	90.5	86.7	88.6	18.0
S3 Failure Precursors	88.4	83.9	86.1	7.4
Macro-F1 (Average)	—	—	89.9	—

As shown in Table 2, the proposed scenario-specific evaluation model demonstrates high classification accuracy across different operational states. The F1 scores for Normal Operation (S0) and Minor Abnormalities (S1) scenarios are 94.5% and 90.4%, respectively, indicating stable recognition performance. For Pre-Failure Scenarios (S3), the F1 score reaches 86.1%, demonstrating the model's strong robustness against rare failure modes. The overall Macro-F1 score is 89.9%, representing an improvement of approximately 3.0 percentage points compared to the baseline model (TCN-only at 86.9%), validating the advantages of the TCN-Transformer fusion architecture. The model effectively addresses the state discrimination requirements of converter equipment under diverse operating conditions and time-varying environments. In simulation tests with 10%-15% Gaussian white noise introduced, the model's Macro-F1 decreased by approximately 3.2 percentage points, while AUC dropped by 2.6 percentage

points. To enhance robustness, noise perturbation augmentation and discriminative loss regularization were incorporated during training. After anti-noise optimization, performance degradation was controlled within 1.5%, significantly improving interference resistance^[13].

B. Diagnostic Performance Metric (AUC)

To quantitatively evaluate the comprehensive diagnostic capability of the constructed model in multi-fault identification tasks for converter equipment, this paper introduces the Area Under the Curve (AUC) as the primary performance metric. AUC reflects the model's overall classification capability across different decision thresholds, with values ranging from $[0,1]$. A higher value indicates greater ability to distinguish between fault and non-fault states. The mathematical formula for AUC is as follows^[14].

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR})d(\text{FPR}) \quad (5)$$

Among these, TPR denotes true positive rate (recall), while FPR represents false positive rate. By calculating TPR and FPR at different decision thresholds, a ROC curve is generated and integrated to obtain the AUC value, which serves as a metric for evaluating the classification performance of faults. Based on the test set data, four primary fault types were selected: IGBT open circuit (OC), capacitor aging (AGE), sensor bias (BIAS), and reduced cooling efficiency (COOL). The diagnostic performance results for each model are summarized in Table 3.

Table 3. Comparison of AUC Metrics for Different Models Across Various Fault Identification Tasks

Model Method	OC Fault	AGE Fault	BIAS Fault	COOL Fault	Average AUC (%)
Rule	86.4	83.7	84.9	85.1	85
TCN-only	91.3	89.8	90.2	89.6	90.2
LSTM-DQN	92.6	91.1	90.8	91.2	91.4
Ours (TCN + Transformer + PPO)	96.1	95.7	94.9	95	95.4

As shown in Table 3, when using TCN+Transformer multi-task learning model in combination with agent decision mechanism, the result shows that the method can recognize well four typical fault types, and the average AUC of the method reaches 95.4%, which is 10.4 percentage points more than traditional method. In terms of the ability of distinguishing high-frequency dynamic feature and gradual aging, the method shows superiority, meanwhile, the method can suppress the interference of noise and cooling anomaly. The final results show that when the model is applied in the maintenance of complex operating conditions, the model can still maintain high diagnostic accuracy and good generalization ability, which can provide a reliable basis for the agent maintenance decision^[15].

The model maintains a high level of diagnostic performance even under compound fault scenarios, such as "IGBT open-circuit combined with cooling failure" and "capacitor aging combined with sensor offset." In the hybrid fault test set, the model consistently achieved an average AUC exceeding 93.2%, indicating its ability to effectively distinguish superimposed anomalous patterns through multi-scale temporal feature

learning. These results demonstrate the model's strong fault identification capability under complex operating conditions.

C. Mean Rank Evaluation

To comprehensively evaluate the overall performance of the proposed intelligent agent model in terms of economic efficiency and reliability, this paper introduces the Operation-Maintenance Benefit (denoted as J) as a quantitative metric. This indicator holistically considers factors such as equipment availability, maintenance costs, and downtime losses to measure the agent's overall benefit level under multi-cycle decision-making. Its definition formula is as follows.

$$J = \sum_{t=1}^T (\lambda_1 A_t - \lambda_2 C_m(a_t) - \lambda_3 C_e(t)) \quad (6)$$

Here, A_t denotes the availability of the equipment at time t , $C_m(a_t)$ represents the cost function of maintenance action a_t , $C_e(t)$ signifies the energy loss caused by downtime, and $\lambda_1, \lambda_2, \lambda_3$ are weighting coefficients used to balance reliability and economic considerations.

To validate the economic efficiency of the agent strategy, we compared it against a traditional rule-based model (Rule), a model relying solely on temporal features (TCN-only), an LSTM-DQN model based on reinforcement learning, and the proposed TCN+Transformer+PPO agent. Experimental results show the profits generated over a 30-day operational cycle, as presented in Table 4.

Table 4. Comprehensive Operational and Maintenance (O&M) Benefits and Related Indicators of Different Models over a 30-Day Cycle

Model Method	Revenue J (10^4 CNY)	Average Availability A (%)	Maintenance Cost C_m (10^4 CNY)	Energy Loss C_e (10^4 CNY)	Cost Reduction Rate ΔC (%)
Rule	112	94.3	68.5	41.2	0
TCN-only	148.3	95.9	63	34.8	8
LSTM-DQN	171.2	96.5	59.7	31.4	12
Ours (TCN + Transformer + PPO)	209	98.2	55.5	27.3	19

As shown in Table 4, when TCN+Transformer+PPO model is used, the maximum profit in 30-day period reaches 2.09 million yuan, which is an increase of 86.6% compared with traditional strategy. In order to achieve the effect of health preservation and cost saving, the model adjusts the number of maintenance and repair strategy according to the running status of equipment. The availability of equipment can reach 98.2%. The cost of maintenance and loss of energy are reduced by 19% and 33% respectively. The results show that the intelligent agent has obvious advantages in economy, stability and reliability, and provides an effective reference for the intelligent O&M of converter equipment.

VI. CONCLUSIONS AND DISCUSSION

Based on empirical research in evaluating certain operational scenarios for converter equipment maintenance and

agent-based operational decision-making model, we can reach the following three conclusions: Firstly, in the scenario recognition, the Macro-F1 score of the model is 89.9%, which is 3 percentage points higher than that of the baseline model. The recognition accuracy and generalization ability of the model in different operating conditions are stronger. Secondly, in the fault diagnosis ability, the average AUC of the model is 95.4%. The average classification ability of the model is stable and reliable, and the classification ability of general typical faults such as open circuit, aging and cooling is satisfied. Thirdly, from the economic perspective, the comprehensive profit of the agent is 2.09 million yuan, the availability rate is 98.2%, and the cost is reduced by 19%. The comprehensive optimization effect of safety, efficiency and cost are satisfied.

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